

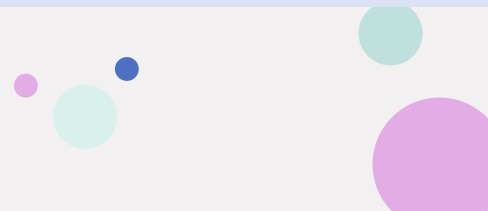


Lecture 5

Hybrid Arguments, More about PRGs

CS409: Introduction to Cryptography

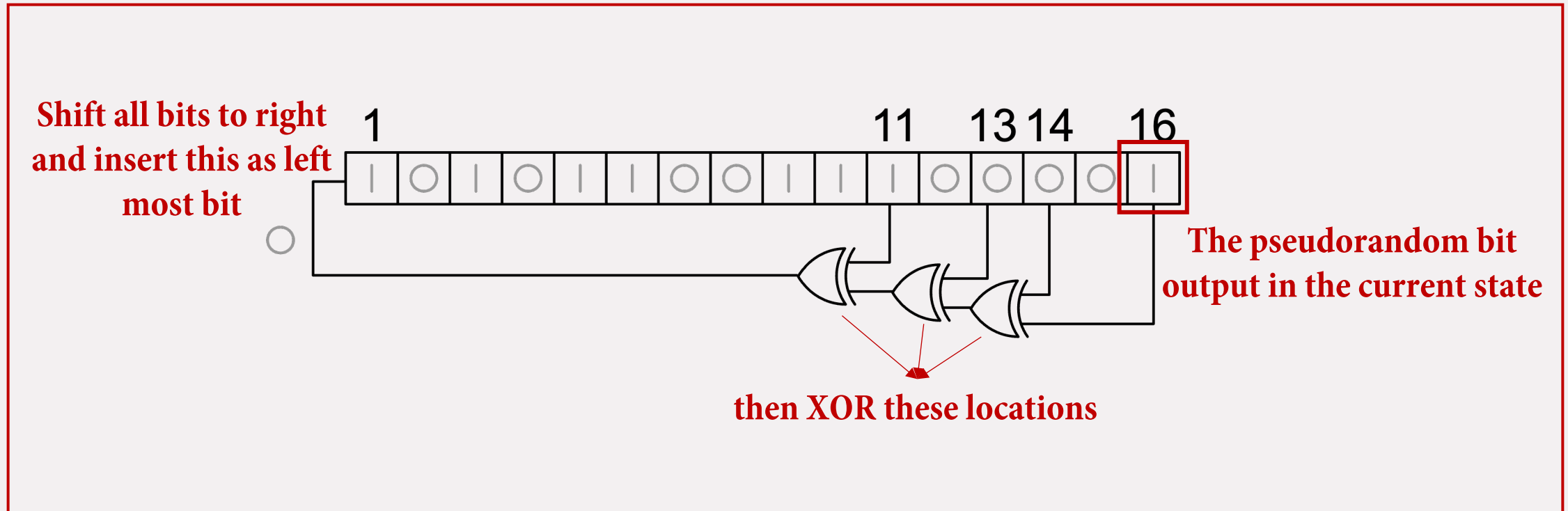
Instructor: Sruthi Sekar



Recap

- We defined two notions for Pseudorandom Generators:
 - Unpredictability
 - Pseudorandomness
- We saw a new proof technique called **security reduction**:
 - Pseudorandomness $\Rightarrow$ Unpredictability
 - Computational Indistinguishability of Computational one-time pad
- Some ideas for building PRGs
 - **RECALL: PRGs do not exist** if
 - you have an unbounded Eve, or
 - if $\mathbf{P} = \mathbf{NP}$

PRGs in the Wild



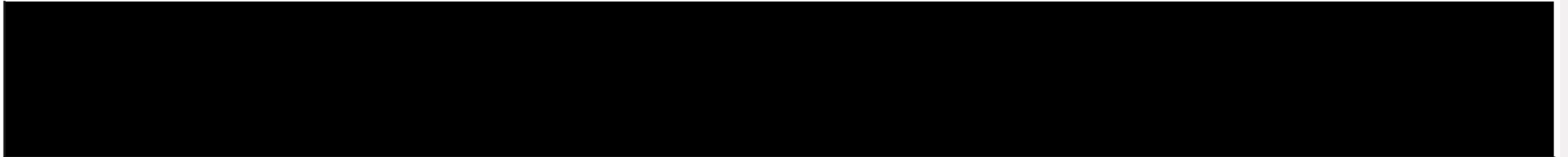
Linear Feedback Shift Register (LFSR)
(how they generate a “pseudorandom bit stream”)

PRGs in the Wild

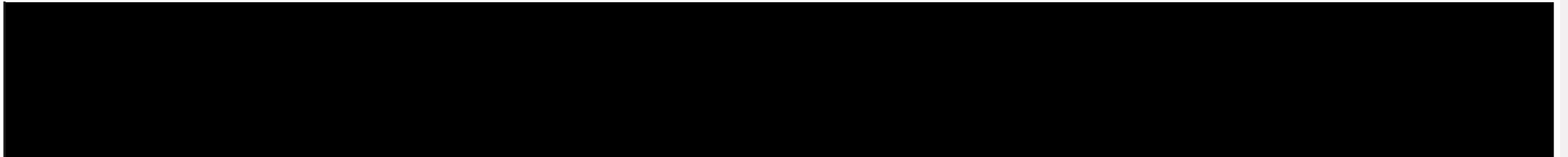
Short cycle (period 5)



Maximal cycle (period 15)



Bad config (period 3)



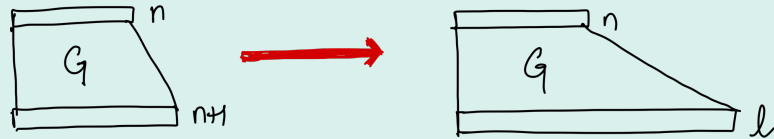
Linear Feedback Shift Register (LFSR)

(how the bits generated by it look like) White: 0 Black: 1

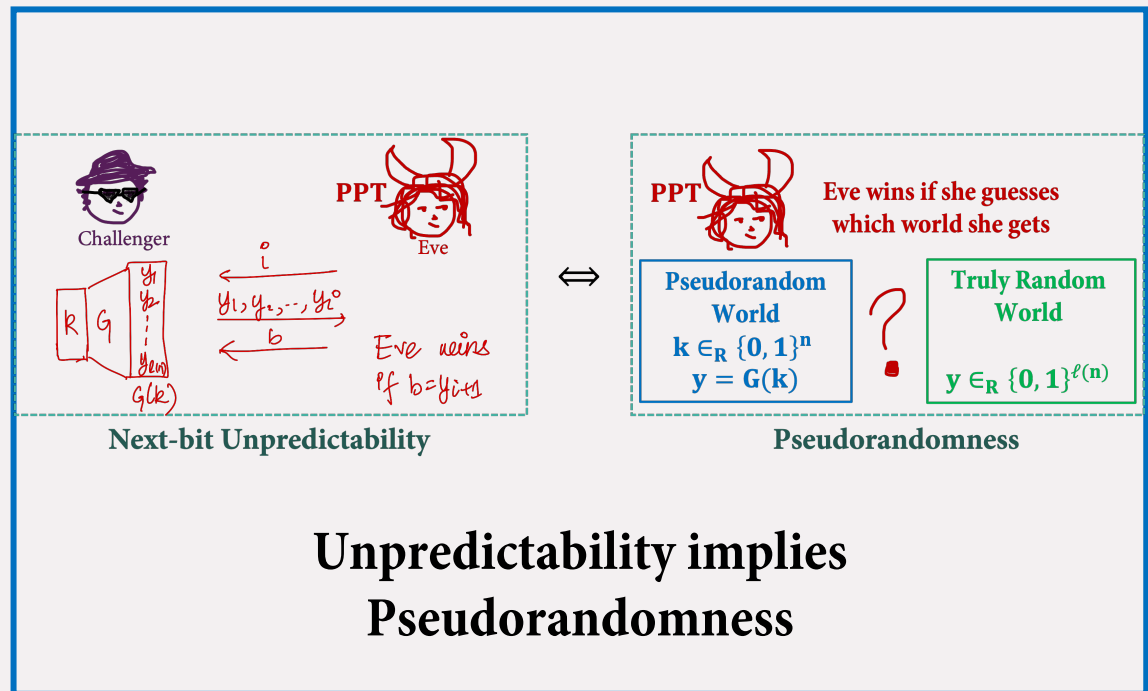
(Predictable $\Rightarrow$ not Pseudorandom, but were used in Satellite TV ciphers in 1990s and early 2000s)

Today

Proof technique called “**hybrid argument**”

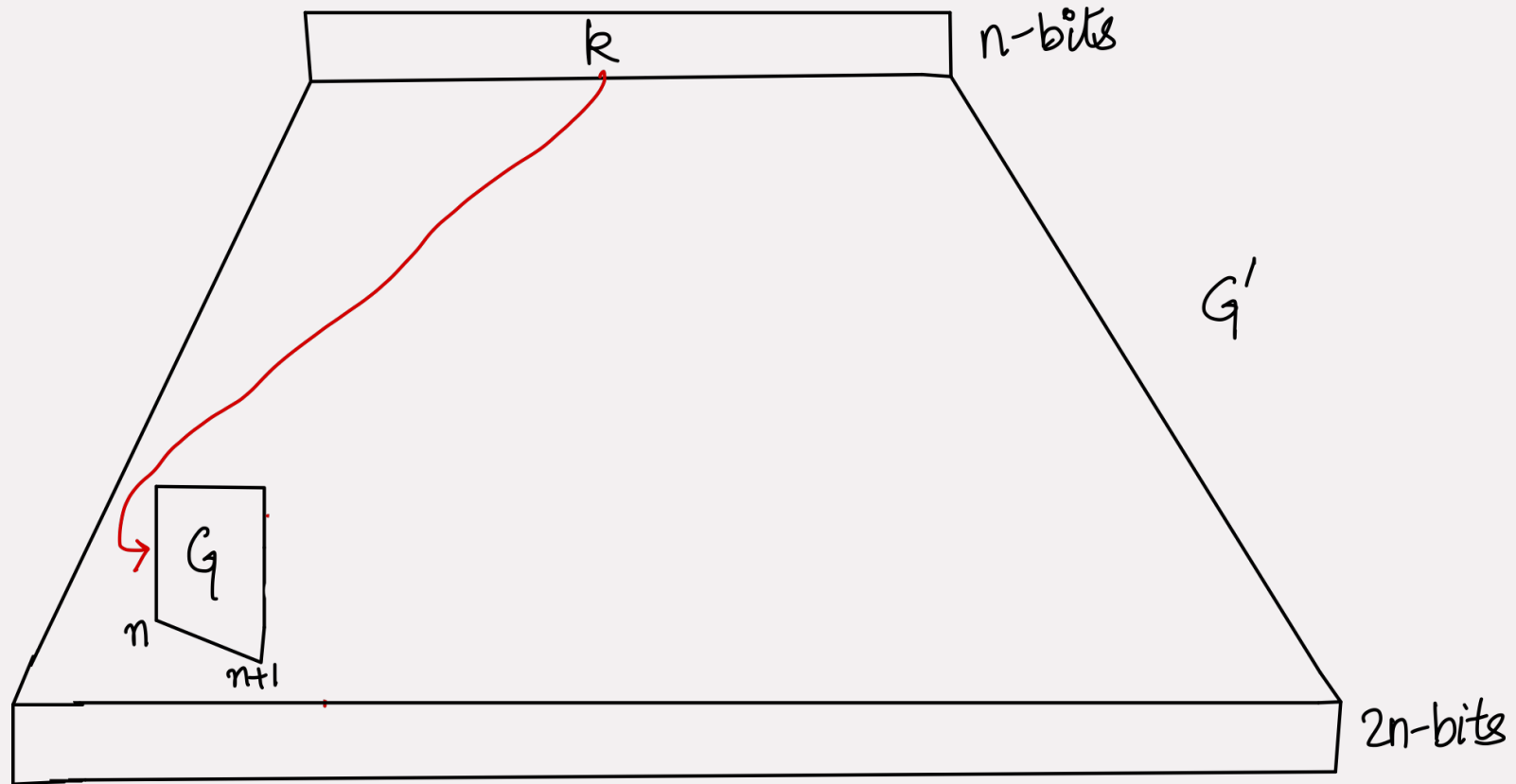


Length-Extension of PRGs



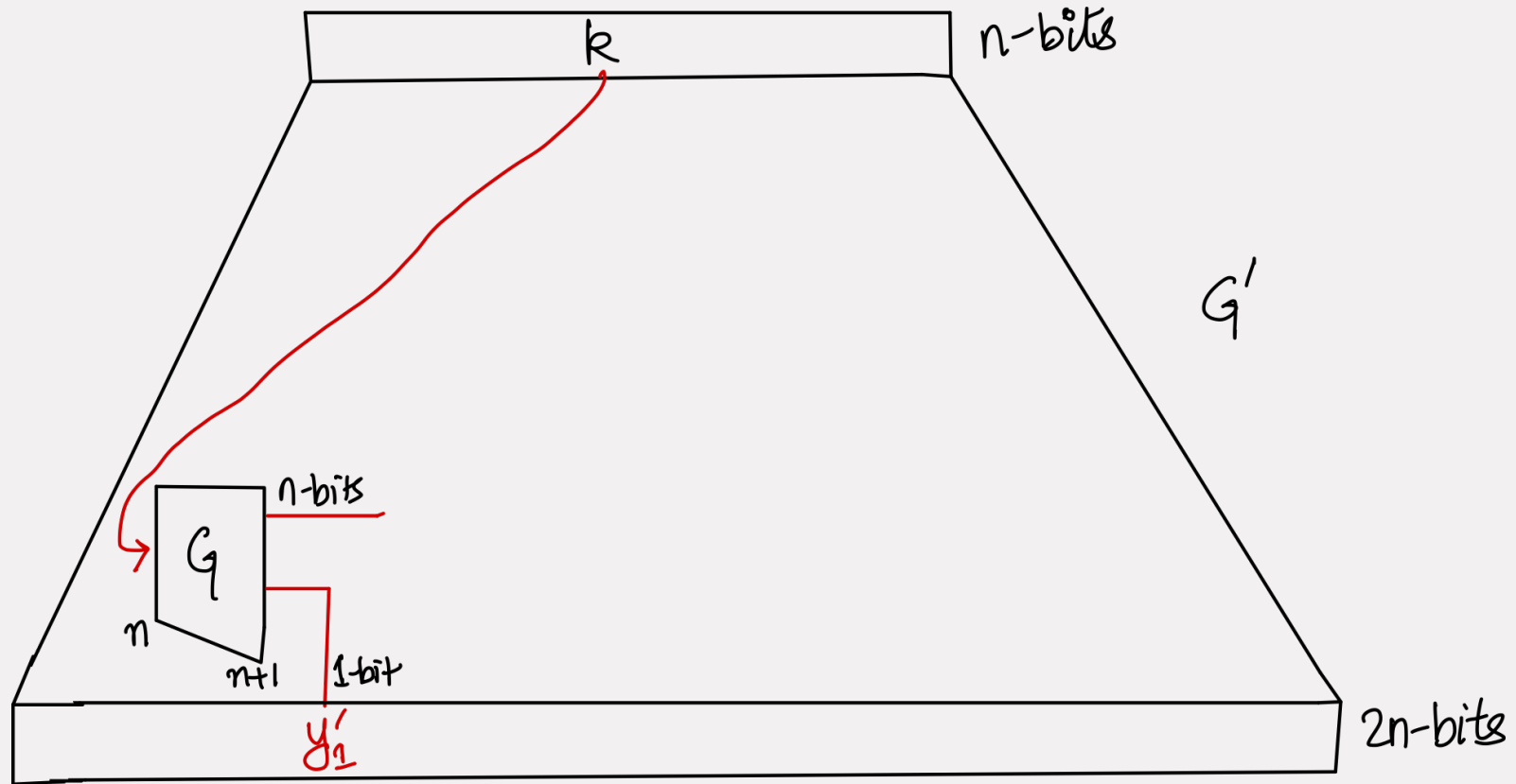
Length Extension of PRGs

GOAL: Given a PRG, G with stretch $n + 1$, can we build a PRG, G' with stretch $2n$



Length Extension of PRGs

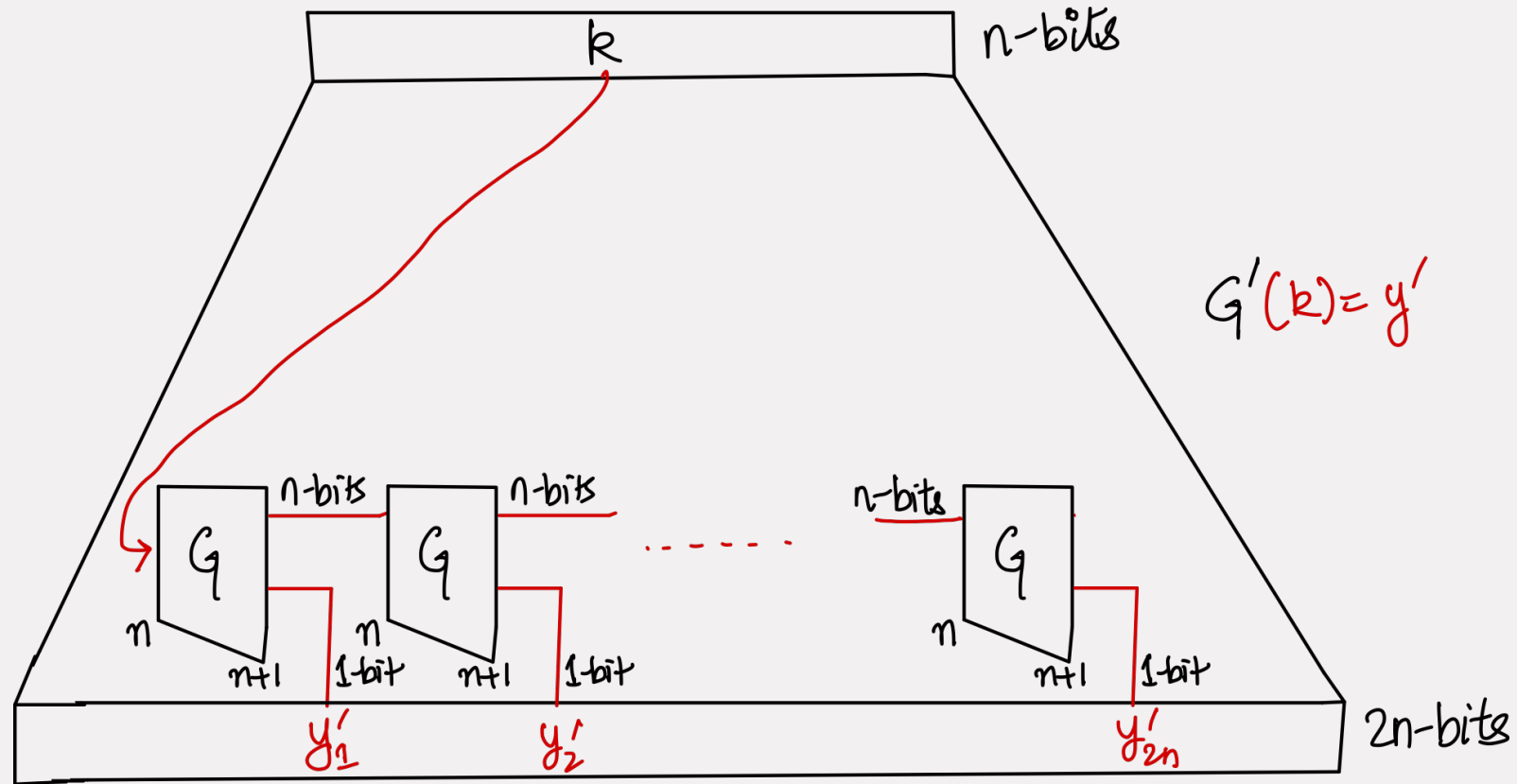
GOAL: Given a PRG, G with stretch $n + 1$, can we build a PRG, G' with stretch $2n$



Construction 1

Length Extension of PRGs

GOAL: Given a PRG, G with stretch $n + 1$, can we build a PRG, G' with stretch $2n$



Construction 1

Length Extension of PRGs

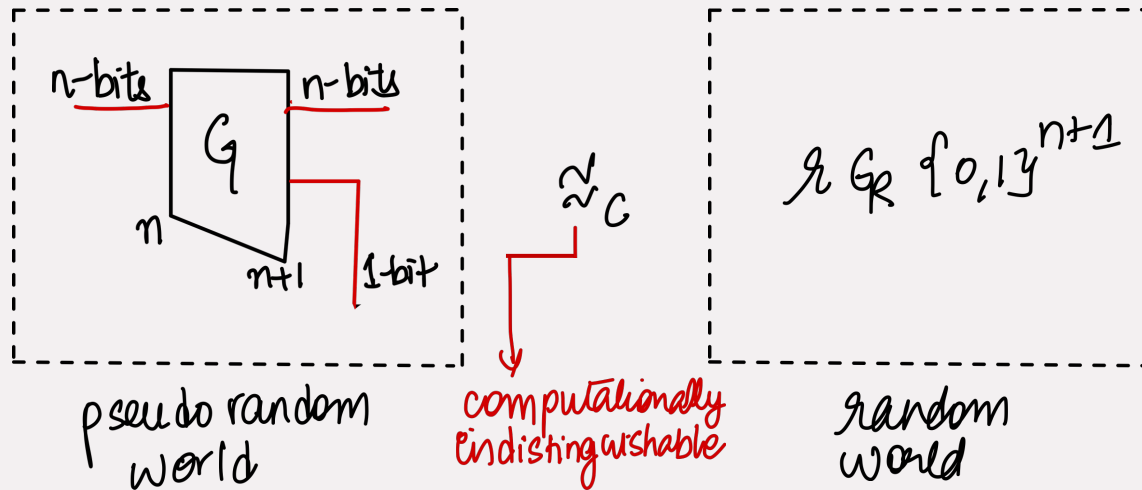
Theorem 1

If G is a pseudorandom generator, then so is G' .



INTUITION

Since G is a PRG



Length Extension of PRGs

Theorem 1

If G is a pseudorandom generator, then so is G' .

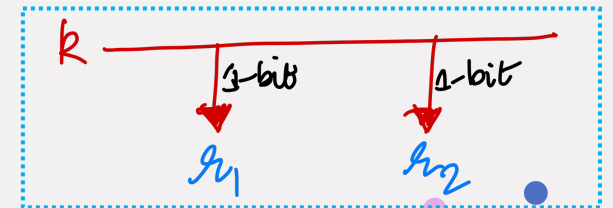
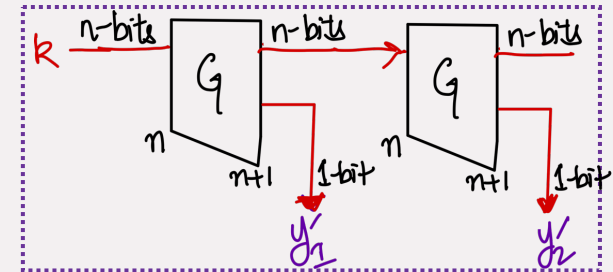


INTUITION Let's understand two iterations first...

Goal is to show that these two worlds are computationally indistinguishable.

H_0 : Pseudorandom World

H_2 : Random World



Length Extension of PRGs

Theorem 1

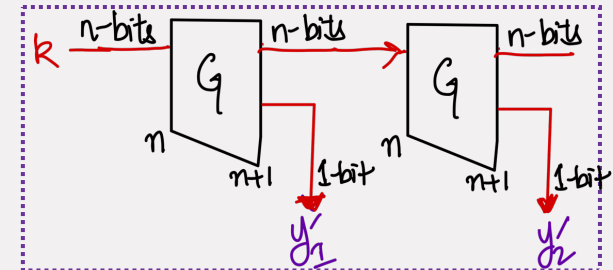
If G is a pseudorandom generator, then so is G' .



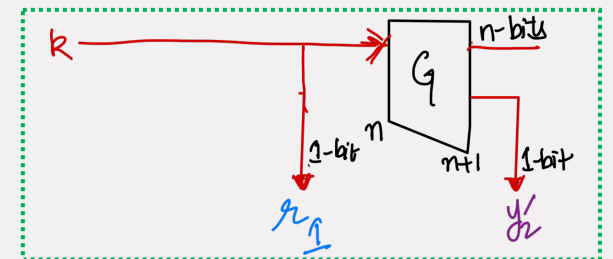
INTUITION Let's understand two iterations first...

Consider a hybrid world between the two of them as an intermediate step!

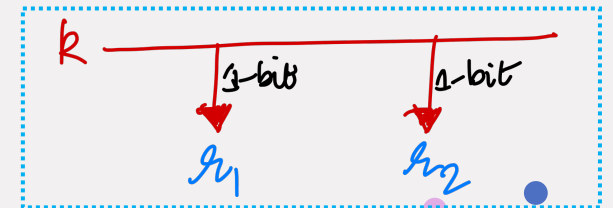
H_0 : Pseudorandom World



H_1 : Hybrid World



H_2 : Random World



Length Extension of PRGs

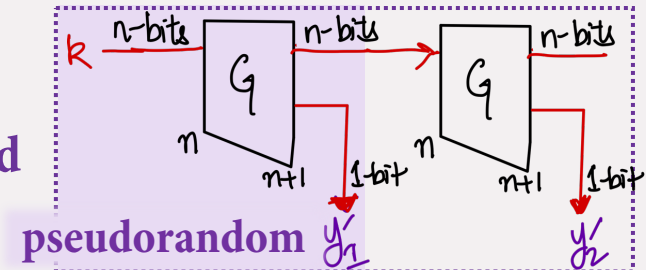
Theorem 1

If G is a pseudorandom generator, then so is G' .



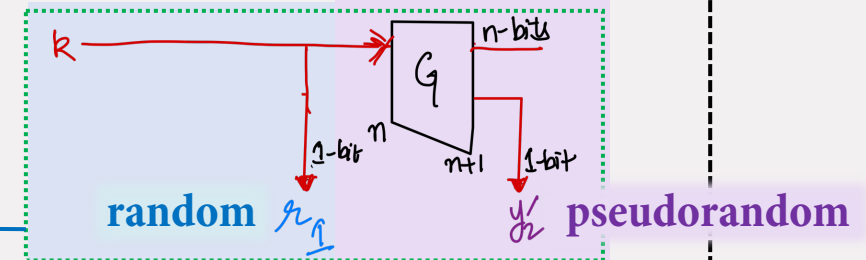
INTUITION Let's understand two iterations first...

H_0 : Pseudorandom World

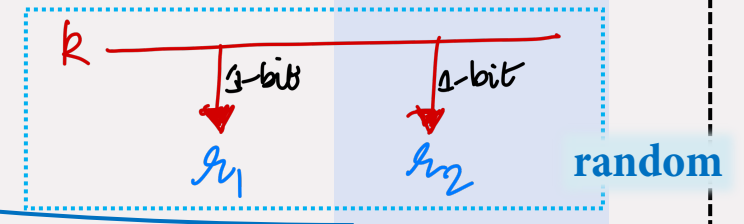


Computationally indistinguishable
since G is a PRG!

H_1 : Hybrid World



H_2 : Random World



Length Extension of PRGs

Theorem 1

If G is a pseudorandom generator, then so is G' .



INTUITION Let's understand two iterations first...

Claim 1: If there exists $\text{Eve}^{G'}$ such that

$$|\Pr[\text{Eve}^{G'}(H_0) = 1] - \Pr[\text{Eve}^{G'}(H_2) = 1]| > \delta$$

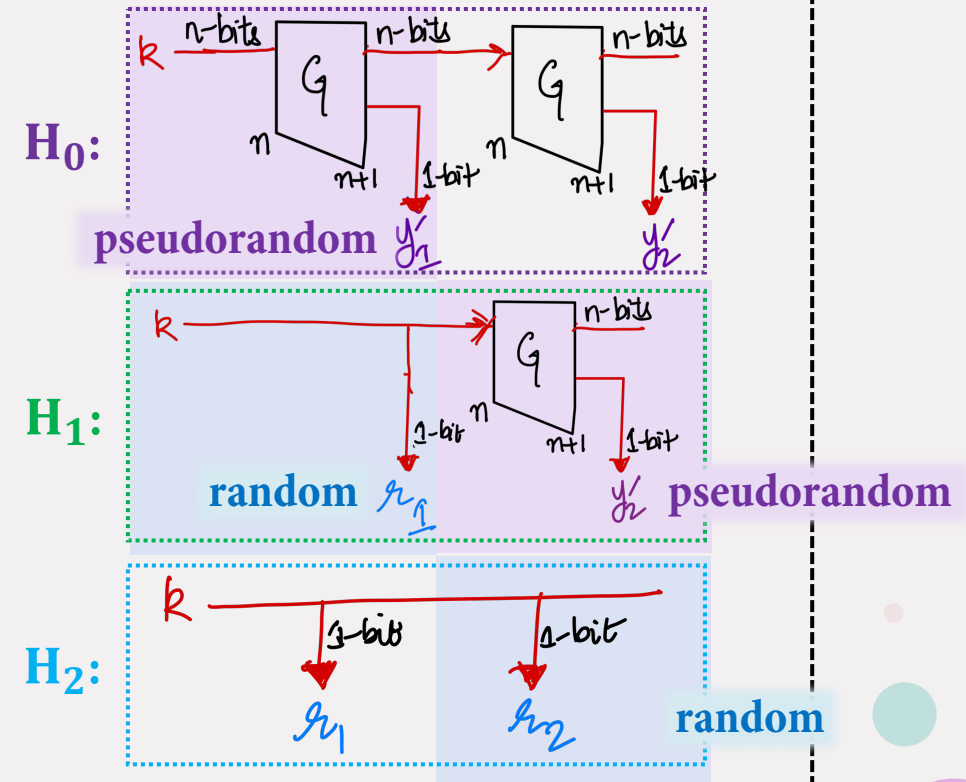
then, $\exists i \in \{0, 1\}$ such that

$$|\Pr[\text{Eve}^{G'}(H_i) = 1] - \Pr[\text{Eve}^{G'}(H_{i+1}) = 1]| > \delta/2$$

Claim 2:

$$|\Pr[\text{Eve}^{G'}(H_i) = 1] - \Pr[\text{Eve}^{G'}(H_{i+1}) = 1]| > \delta/2 \\ \Rightarrow \text{A distinguisher } \text{Eve}^G \text{ for } G$$

Claim 1 & Claim 2 $\Rightarrow$ Theorem 1



Length Extension of PRGs

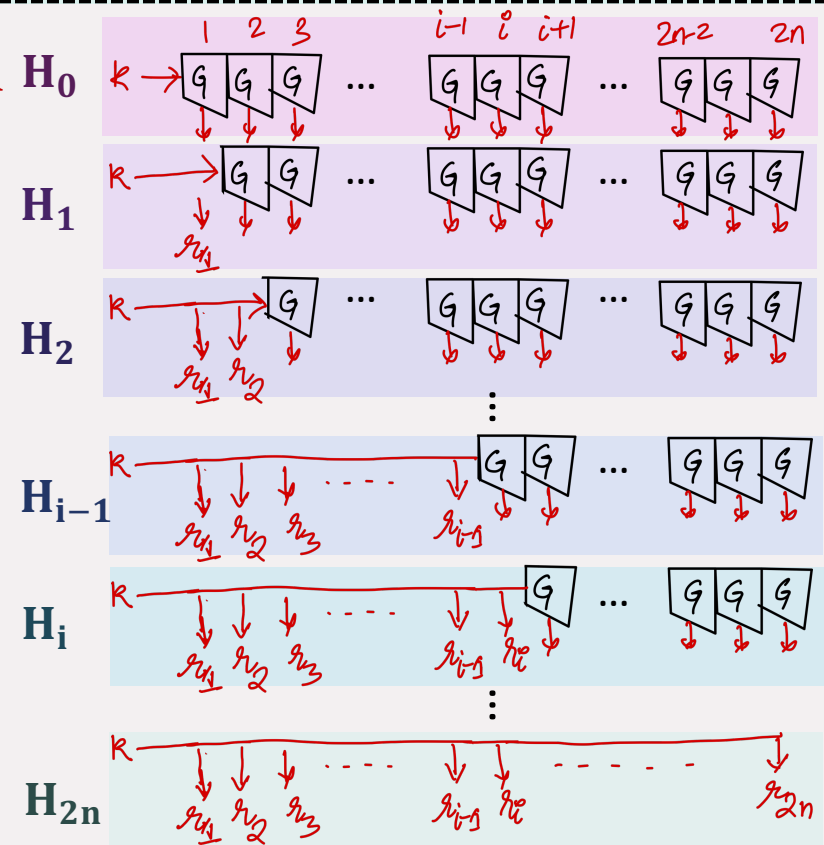
Theorem 1

If G is a pseudorandom generator, then so is G' .



INTUITION (All hybrids)

pseudorandom H_0



Length Extension of PRGs

Theorem 1

If G is a pseudorandom generator, then so is G' .



INTUITION (All hybrids)

Each pair of hybrids look computationally indistinguishable since G is a PRG

pseudorandom H_0

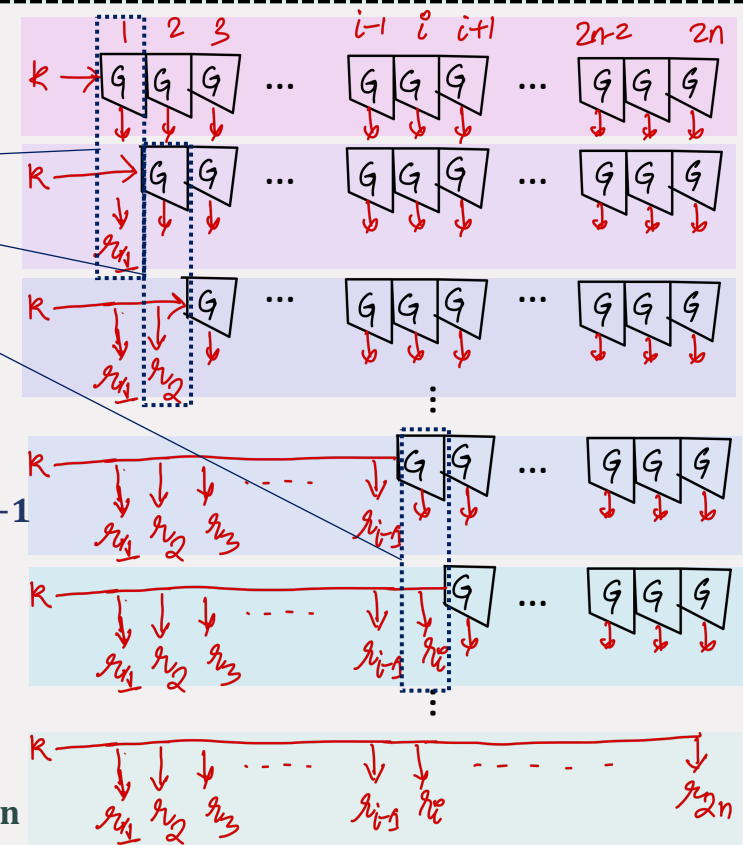
H_1

H_2

H_{i-1}

H_i

random H_{2n}



Length Extension of PRGs

Theorem 1

If G is a pseudorandom generator, then so is G' .

Proof: Suppose $\exists \text{Eve}^{G'}$ that breaks pseudorandomness of G'

Claim 1: If there exists $\text{Eve}^{G'}$ such that
 $|\Pr[\text{Eve}^{G'}(H_0) = 1] - \Pr[\text{Eve}^{G'}(H_{2n}) = 1]| > \delta$
then, $\exists i \in \{0, 1, \dots, 2n - 1\}$ such that
 $|\Pr[\text{Eve}^{G'}(H_i) = 1] - \Pr[\text{Eve}^{G'}(H_{i+1}) = 1]| > \delta/2n$

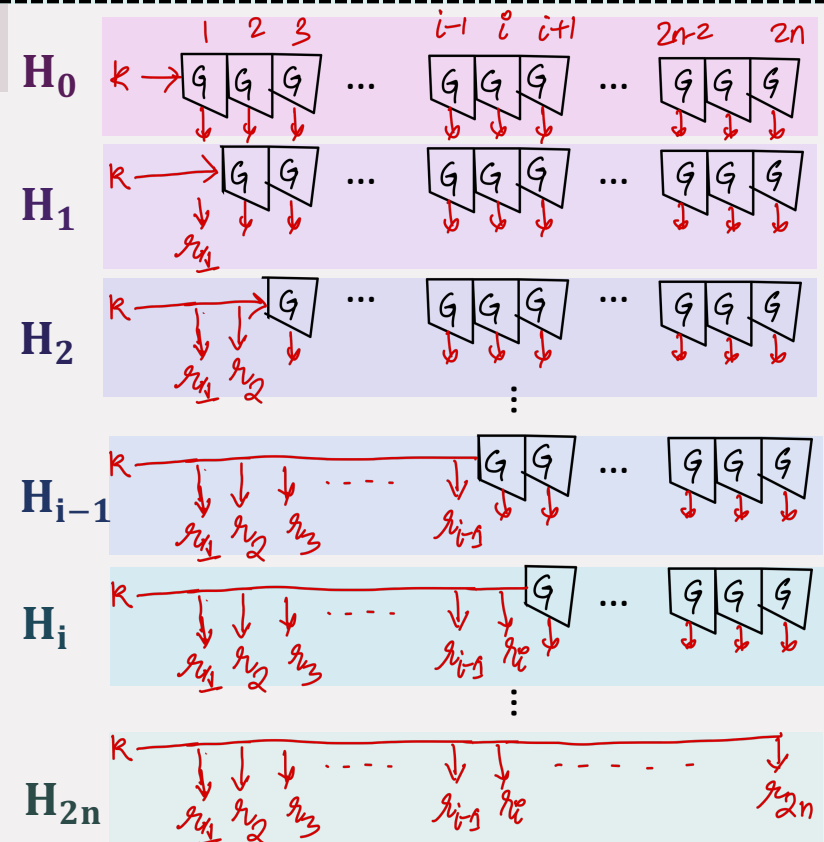


This claim will be implicitly used in our proof!

Eve^G wants to pick this “ i ” and use $\text{Eve}^{G'}$ to win her game!



Eve^G doesn't know which i ,
so she will just pick one at random!



Length Extension of PRGs

Theorem 1

If G is a pseudorandom generator, then so is G' .

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Claim 1: If there exists $\text{Eve}^{G'}$ such that $|\Pr[\text{Eve}^{G'}(H_0) = 1] - \Pr[\text{Eve}^{G'}(H_{2n}) = 1]| > \delta$
then, $\exists i \in \{0, 1, \dots, 2n - 1\}$ such that $|\Pr[\text{Eve}^{G'}(H_i) = 1] - \Pr[\text{Eve}^{G'}(H_{i+1}) = 1]| > \delta/2n$

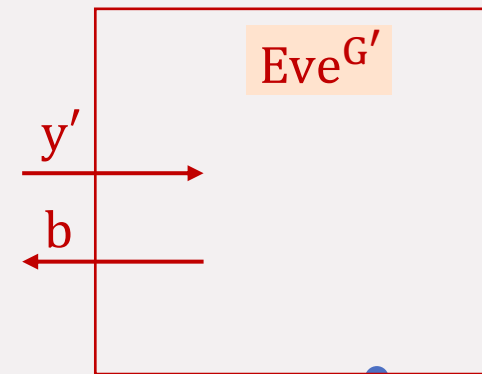
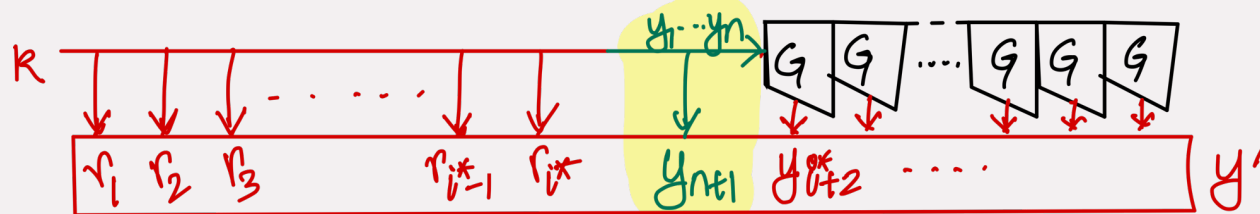
Eve^G

$$\Pr[i^* = i] = \frac{1}{2n}$$

Input: $y \in \{0,1\}^{n+1}$

- Picks $i^* \in_R \{0, \dots, 2n - 1\}$
- Computes challenge for $\text{Eve}^{G'}$ as:

Outputs: b



Length Extension of PRGs

Theorem 1

If G is a pseudorandom generator, then so is G' .

Proof: Suppose $\exists \text{Eve}^{G'}$ that breaks pseudorandomness of G'

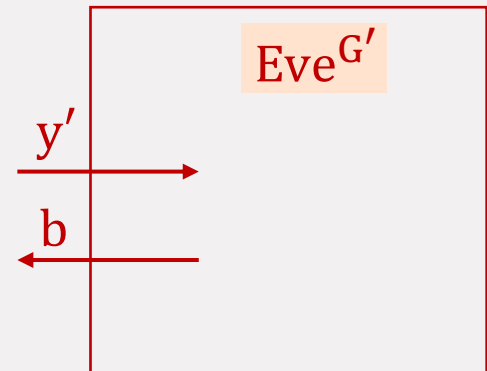
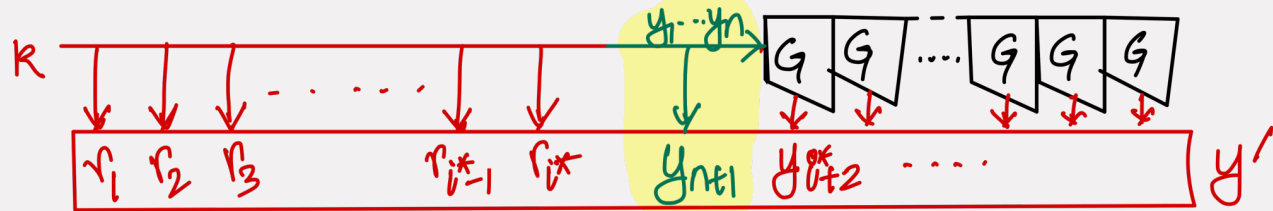
Combining ideas to complete the proof!

Eve^G

Input: $y \in \{0,1\}^{n+1}$

- Picks $i^* \in_R \{0, \dots, 2n-1\}$
- Computes challenge for $\text{Eve}^{G'}$ as:

Outputs: b



$$\begin{aligned}
 & |\Pr[\text{Eve}^G(G(k)) = 1] - \Pr[\text{Eve}^G(r) = 1]| \\
 &= |\Pr[\text{Eve}^{G'}(H_{i^*}) = 1] - \Pr[\text{Eve}^{G'}(H_{i^*+1}) = 1]| = \frac{1}{2n} \left| \sum_{i=0}^{2n-1} \Pr[\text{Eve}^{G'}(H_i) = 1] - \sum_{i=0}^{2n-1} \Pr[\text{Eve}^{G'}(H_{i+1}) = 1] \right| \\
 &= \frac{1}{2n} |\Pr[\text{Eve}^{G'}(H_0) = 1] - \Pr[\text{Eve}^{G'}(H_{2n}) = 1]| > \frac{\delta}{2n} \blacksquare
 \end{aligned}$$

Can we keep stretching?

- [Construction 1](#) can be generalized to work for a $p(n)$ -extension, where p is a polynomial. **WHY?**
- Can you get an exponential stretch?
- What is the loss here?

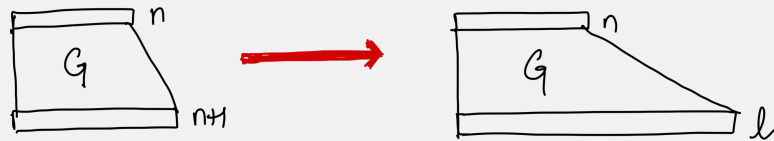
When $\text{Eve}^{G'}$ distinguishes with probability $\delta(n)$ then Eve^G only distinguishes with probability $\frac{\delta(n)}{p(n)}$
 $\Rightarrow$ More you stretch, greater is the security loss!



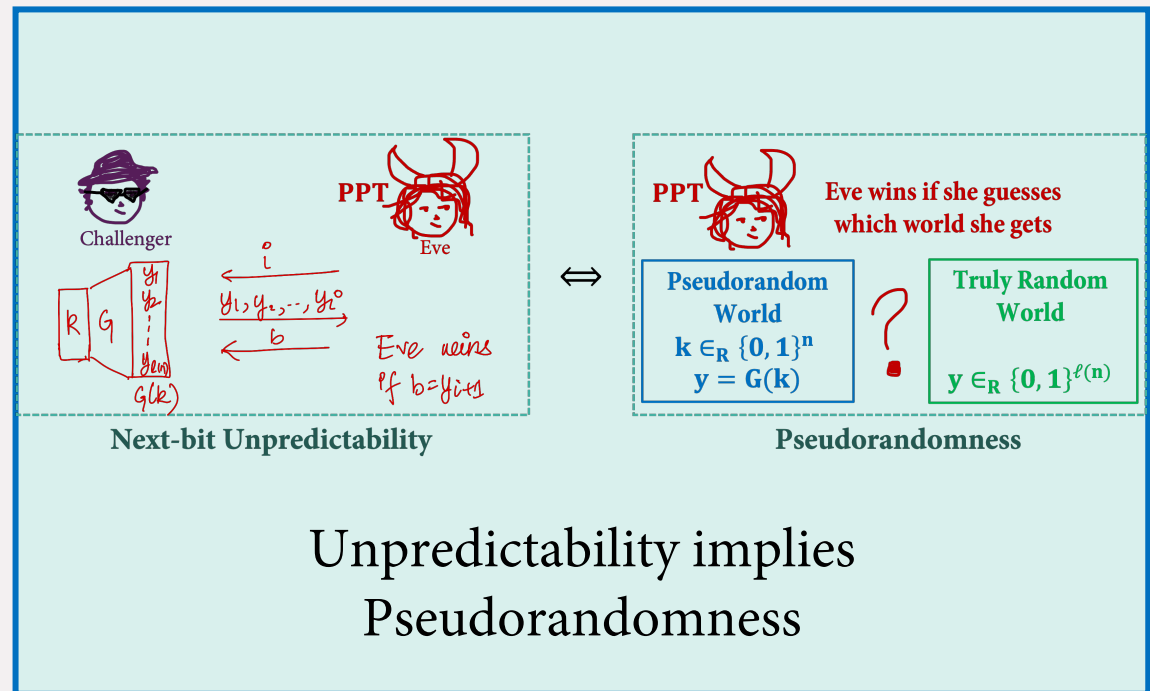
Is there a better construction that doesn't result in this much loss?

Today

Proof technique called “**hybrid argument**”

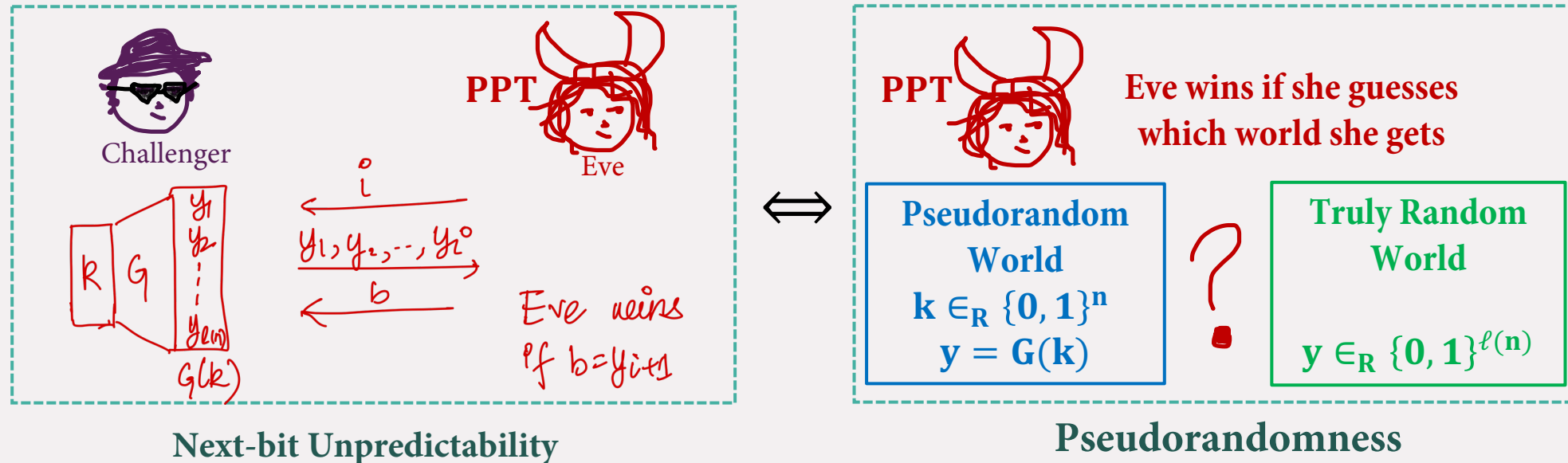


Length-Extension of PRGs



Pseudorandom Generators

Recall: Equivalence of the Definitions



Theorem (from Lecture 4)

Let ℓ be a polynomial and let G be a deterministic polynomial-time algorithm such that for any n , and any $k \in \{0, 1\}^n$, $G(k) \in \{0, 1\}^{\ell(n)}$, where $\ell(n) > n$ (expanding).

G is pseudorandom if and only if G is next-bit unpredictable.

Pseudorandom Generators

Unpredictability $\Rightarrow$ Pseudorandomness

Theorem 2

If G is next-bit unpredictable, then G is pseudorandom.

Proof Sketch: Assume to the contrary that G is not pseudorandom. **To prove:** G is next-bit predictable

$\exists \text{Eve}^{\text{PR}}$ such that

$$\left| \Pr_{r \in_R \{0,1\}^{\ell(n)}} [\text{Eve}^{\text{PR}}(H_0 = r) = 1] - \Pr_{k \in_R \{0,1\}^n} [\text{Eve}^{\text{PR}}(H_{\ell(n)} = G(k)) = 1] \right| > \delta(n)$$



STEP 1: Hybrid Argument



STEP 2: From distinguishing to predicting

Pseudorandom Generators

Unpredictability $\Rightarrow$ Pseudorandomness

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Proof Sketch: Assume to the contrary that G is not pseudorandom. **To prove:** G is next-bit predictable

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Claim 1: $\exists i \in \{0, \dots, \ell(n) - 1\}$ such that
 $|\Pr[\text{Eve}^{\text{PR}}(H_i) = 1] - \Pr[\text{Eve}^{\text{PR}}(H_{i+1}) = 1]| > \delta(n)/\ell(n)$

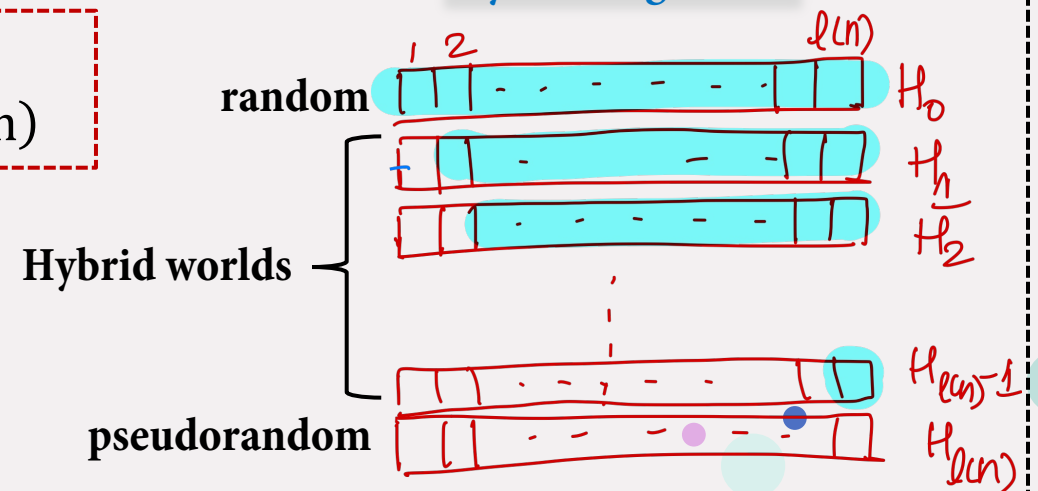


$\text{Eve}^{\text{predictor}}$ wants to pick this “ i ” and use Eve^{PR} to win her game!



$\text{Eve}^{\text{predictor}}$ doesn't know which i , so she will just pick one at random!

Hybrid Argument



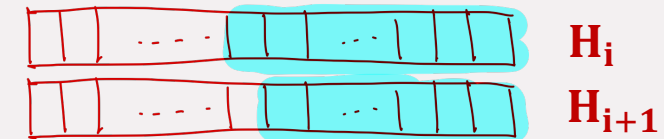
Pseudorandom Generators

Unpredictability $\Rightarrow$ Pseudorandomness

Theorem 2

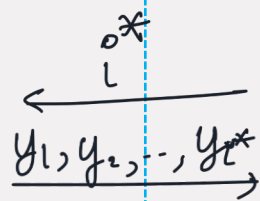
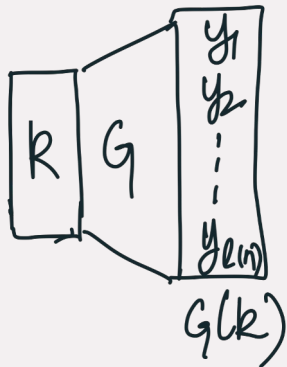
If G is next-bit unpredictable, then G is pseudorandom.

Claim 1: $\exists i \in \{0, \dots, \ell(n) - 1\}$ such that
 $|\Pr[\text{Eve}^{\text{PR}}(\mathbf{H}_i) = 1] - \Pr[\text{Eve}^{\text{PR}}(\mathbf{H}_{i+1}) = 1]| > \delta(n)/\ell(n)$



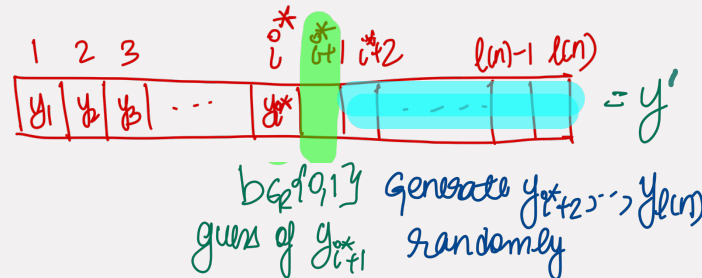
Eve^{predictor}

Challenger^{Unpred}



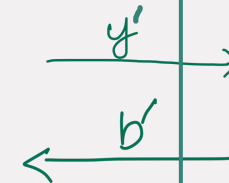
b , if $b' = 1$
 $\bar{b}$, if $b' = 0$

$i^* \in \{0, 1, \dots, \ell(n) - 1\}$



“acts as challenger”

Eve^{PR}



Pseudorandom Generators

Unpredictability $\Rightarrow$ Pseudorandomness

Theorem 2

If G is next-bit unpredictable, then G is pseudorandom.

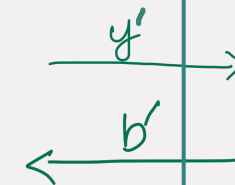
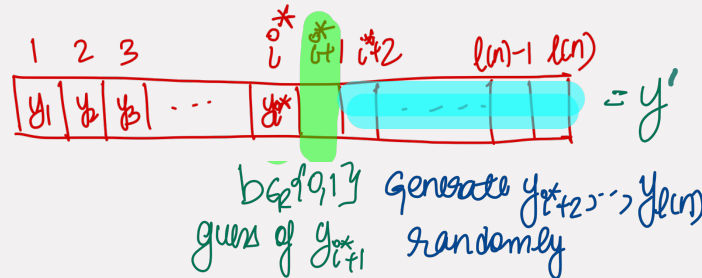
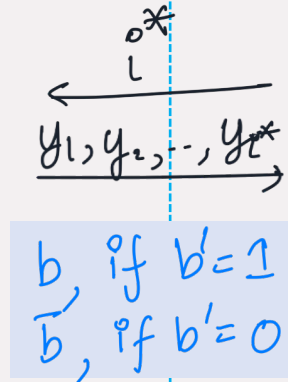
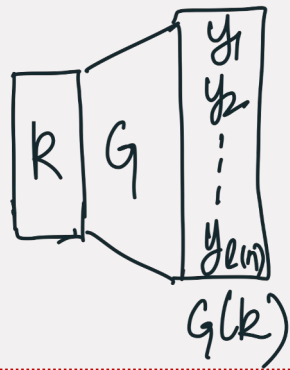
$$\Pr[i^* = i] = 1/\ell(n)$$

Eve^{predictor}

Challenger^{Unpred}

“acts as challenger”

Eve^{PR}



Eve^{PR} outputs 1 more often given a pseudorandom $(i^* + 1)$ th bit than a random $(i^* + 1)$ th bit

Theorem 2

$$\Pr[\mathbf{i}^* = \mathbf{i}] = 1/\ell(\mathbf{n})$$
$$-i^*_{\mathbb{R}} [0, 1, \dots, l(n) - 1]$$
Eve^{PR}

Full proof (if interested):  **Ch 3, Foundations of Cryptography, Volume I, Oded Goldreich**

Summary + Next Class

- **SUMMARY**

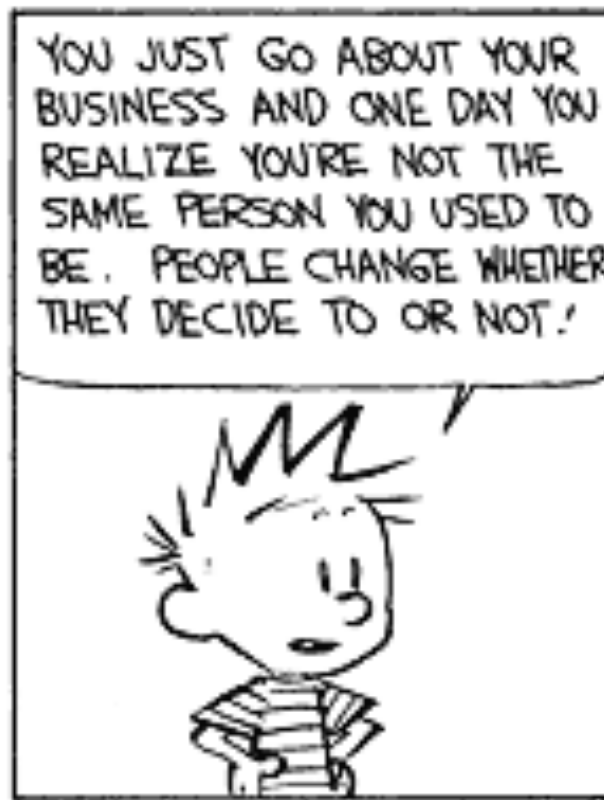
- What have we seen so far?
PRGs (two notions), Extending PRGs, Hybrid Argument
- Computational Indistinguishability of Computational one-time pad
(showed how to overcome the key length drawback of perfect security)

- **NEXT CLASS**

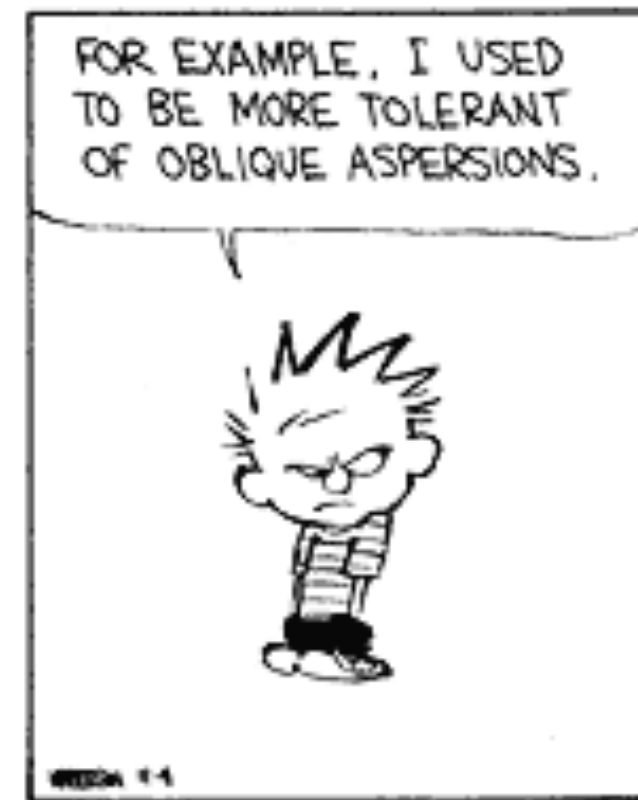
- What about reusing the same key? (overcoming the second drawback of perfect security)
- What about a stronger Eve? (CPA Secure Encryption)
- How do we build stronger schemes? (Pseudorandom Functions, Block ciphers)

Reference for today's proofs

 [Ch 3, Foundations of Cryptography, Volume I, Oded Goldreich](#)



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THANK YOU